

Destabilase, an enzyme found in the saliva of leeches, exhibits at least two activities: isopeptidase activity and glycosidase activity. Isopeptidase activity is believed to prevent blood clotting by hydrolyzing the isopeptide bond between the ϵ -amino group of lysine and the γ -carboxyl group of glutamate in fibrin monomers. The enzyme's glycosidase activity digests sugars found in the peptidoglycans of bacterial cell walls and may play a role in innate immunity.

Two isoforms of destabilase were tagged with a polyhistidine tail and recombinantly expressed in *E. coli* cells. Both isoforms were then purified by affinity chromatography and eluted by imidazole, which competes with histidine-tagged proteins for column binding. Each isoform required 300 mM imidazole to elute efficiently. Then, 5 μ L of each eluted enzyme was provided with saturating levels of substrate and assessed for isopeptidase and glycosidase activity at various pH levels, which allowed the optimal pH of enzymatic activities for each isoform to be identified (Figure 1). Additional studies showed that at optimal pH, both isoforms had approximately the same k_{cat} values, but isoform 2 had a lower K_M than isoform 1 for both enzymatic activities.

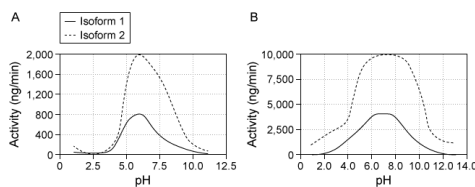


Figure 1 Isopeptidase activity (A) and glycosidase activity (B) of 5 μ L of destabilase isoforms 1 and 2 at various pH values

Isoform 2 was further characterized by mutating specific residues and measuring enzymatic activities at optimal pH. The results are shown in Figure 2 as a percentage of the activity observed in wild-type enzyme.

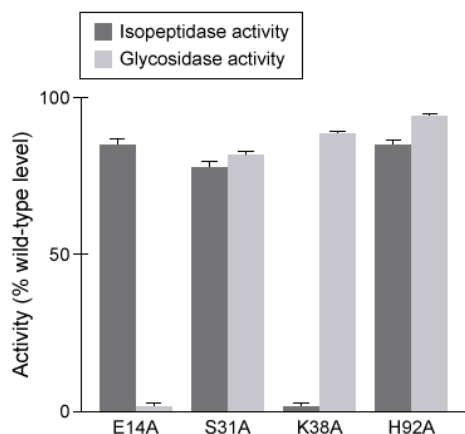


Figure 2 Isopeptidase and glycosidase activity of isoform 2 of destabilase with various mutations

Given that blood pH is 7.4, what is the approximate isopeptidase activity level of isoform 1 in blood expressed as a percentage of optimal activity?

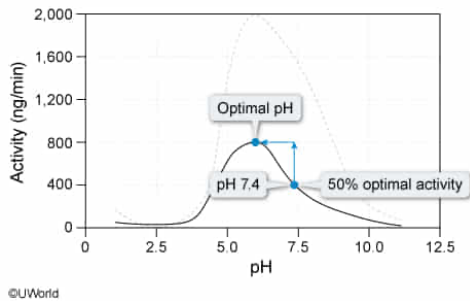
- ☐ A. 40% [20%]
- ☒ B. 50% [59%]
- ☐ C. 80% [17%]
- ☐ D. 100% [2%]

Correct

60% answered correctly

Explanation:

Explanation:



Enzymes consist of several amino acids linked together by peptide bonds. Some amino acids contain side chains that can be protonated or deprotonated depending on the ambient pH. As the protonation state of a side chain changes, its ability to stabilize the enzyme's structure or to interact with substrates changes as well. For this reason, an enzyme's activity typically varies with pH, and each enzyme has an optimal pH, which must be determined experimentally.

Figure 1 shows that for isopeptidase activity, both isoforms of destabilase have an optimal pH of approximately 6. Isoform 1 converts substrate to product at a rate of ~800 ng/min at optimal pH. As pH increases above 6, isopeptidase activity decreases. The graph shows that at the pH of blood (7.4), the isopeptidase activity diminishes to ~400 ng/min. Therefore, at pH 7.4, isoform 1 exhibits isopeptidase activity that is 50% of its optimal activity (400/800).

(Choice A) Figure 1A shows that at optimal pH, isoform 1 has approximately 40% (800/2,000) of the activity of isoform 2 at the same pH. However, the question asks how isoform 1 compares to *itself* as the pH changes from optimal pH (6) to physiological blood pH (7.4).

(Choice C) Figure 1A shows that isoform 2 has approximately 80% of its optimal activity at pH 7.4 (1,600/2,000), but the question asks about isoform 1.

(Choice D) Isoform 1 of destabilase exhibits 100% of its optimal glycosidase activity at a pH of 7.4 (Figure 1B), but the question asks for isopeptidase activity.

Educational objective:

Enzyme activity is dependent on the ambient pH. Changes in pH can alter the protonation states of some amino acid side chains, which can alter their ability to stabilize enzyme structure and interact with substrates. The optimal pH of an enzyme must be determined experimentally.

Time spent: 0 sec

QID: 402745

Subject:
Biochemistry

System:
Enzymes

Destabilase, an enzyme found in the saliva of leeches, exhibits at least two activities: isopeptidase activity and glycosidase activity. Isopeptidase activity is believed to prevent blood clotting by hydrolyzing the isopeptide bond between the ϵ -amino group of lysine and the γ -carboxyl group of glutamate in fibrin monomers. The enzyme's glycosidase activity digests sugars found in the peptidoglycans of bacterial cell walls and may play a role in innate immunity.

Two isoforms of destabilase were tagged with a polyhistidine tail and recombinantly expressed in *E. coli* cells. Both isoforms were then purified by affinity chromatography and eluted by imidazole, which competes with histidine-tagged proteins for column binding. Each isoform required 300 mM imidazole to elute efficiently. Then, 5 μ L of each eluted enzyme was provided with saturating levels of substrate and assessed for isopeptidase and glycosidase activity at various pH levels, which allowed the optimal pH of enzymatic activities for each isoform to be identified (Figure 1). Additional studies showed that at optimal pH, both isoforms had approximately the same k_{cat} values, but isoform 2 had a lower K_M than isoform 1 for both enzymatic activities.

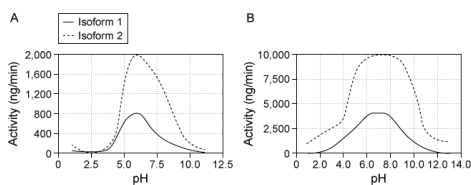


Figure 1 Isopeptidase activity (A) and glycosidase activity (B) of 5 μ L of destabilase isoforms 1 and 2 at various pH values

Isoform 2 was further characterized by mutating specific residues and measuring enzymatic activities at optimal pH. The results are shown in Figure 2 as a percentage of the activity observed in wild-type enzyme.

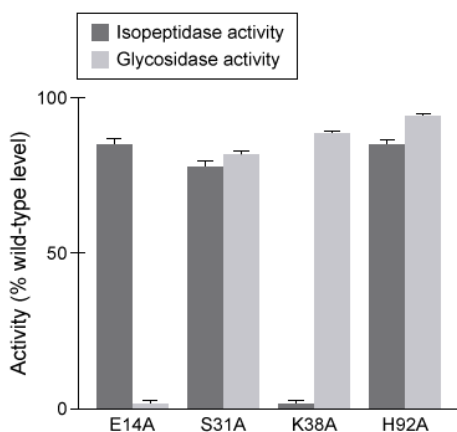
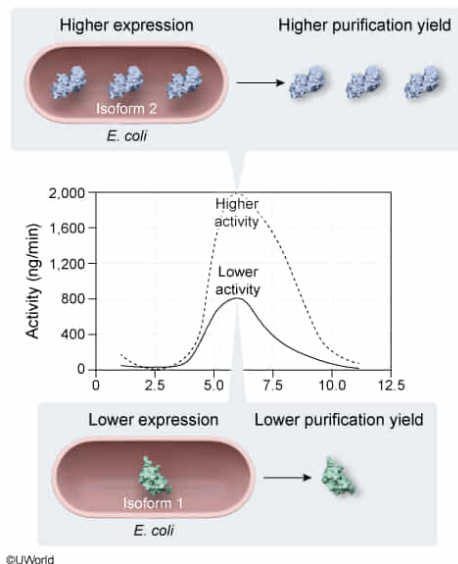


Figure 2 Isopeptidase and glycosidase activity of isoform 2 of destabilase with various mutations

Both isoforms of destabilase have approximately the same k_{cat} values for isopeptidase activity at optimal pH, but Figure 1 shows that when 5 μ L of each purified enzyme was provided with saturating levels of substrate, the isoforms had different levels of activity. What factor could explain this apparent discrepancy?

- ☐ A. Isoform 2 had less activity than isoform 1 because isoform 2 did not operate at V_{max} under saturating conditions. [8%]
- ☐ B. The 5- μ L sample of isoform 1 had a higher enzyme concentration than the isoform 2 sample because the affinity column bound isoform 1 more tightly. [18%]
- ☐ C. Isoform 1 had less activity than isoform 2 because isoform 1 became denatured at pH 6. [13%]
- ☒ D. The 5- μ L sample of isoform 2 had a higher enzyme concentration than the isoform 1 sample because *E. coli* expressed isoform 2 at higher levels. [58%]

Explanation:

An enzyme's **turnover number** k_{cat} represents the number of **substrate molecules converted** to product **per second** by each enzyme in solution under saturating conditions. The **maximum velocity** V_{max} of an enzyme-catalyzed reaction is k_{cat} **multiplied by the enzyme concentration** $[E]$. Therefore, enzymes at a **higher concentration** in solution exhibit a **higher** V_{max} than enzymes with the same k_{cat} value present at a **lower concentration**.

The passage states that isoforms 1 and 2 of destabilase have approximately the same k_{cat} values for both isopeptidase and glycosidase activities at optimal pH, yet Figure 1 shows that 5 μL of isoform 2 has substantially higher activity than the same volume of isoform 1.

The fact that equal *volumes* of each enzyme were used does not necessarily mean that both 5- μL samples had the same enzyme *concentration* after purification. If the bacteria **expressed isoform 2 at higher levels** than isoform 1, this would result in a greater enzyme concentration in the 5- μL sample of isoform 2 (and therefore a higher V_{max}) than in the 5- μL sample of isoform 1. Therefore, it is likely that the bacteria expressed isoform 2 at higher levels than they expressed isoform 1.

(Choice A) Under **saturating conditions**, an enzyme effectively operates at V_{max} . Because both isoforms were saturated, *both* were essentially operating at V_{max} at every pH tested. In addition, isoform 2 had *more* activity than isoform 1.

(Choice B) Both isoforms eluted at the same imidazole concentration (300 mM), indicating that they both have the *same affinity* for the column. In addition, isoform 1 had less activity than isoform 2, indicating a *lower* enzyme concentration.

(Choice C) Isoforms 1 and 2 have approximately the same optimal pH (~6) for isopeptidase activity. If isoform 1 were denatured at pH 6, it would be inactive, and pH 6 would *not* be its optimal pH.

Educational objective:

An enzyme's turnover number k_{cat} is the number of reactions each enzyme catalyzes per second under saturating conditions. The maximum velocity V_{max} of an enzyme-catalyzed reaction is equal to k_{cat} multiplied by the concentration of enzyme $[E]$. Two enzymes with similar k_{cat} values may have different V_{max} values if one of the enzymes is present at a higher concentration than the other.

Destabilase, an enzyme found in the saliva of leeches, exhibits at least two activities: isopeptidase activity and glycosidase activity. Isopeptidase activity is believed to prevent blood clotting by hydrolyzing the isopeptide bond between the ϵ -amino group of lysine and the γ -carboxyl group of glutamate in fibrin monomers. The enzyme's glycosidase activity digests sugars found in the peptidoglycans of bacterial cell walls and may play a role in innate immunity.

Two isoforms of destabilase were tagged with a polyhistidine tail and recombinantly expressed in *E. coli* cells. Both isoforms were then purified by affinity chromatography and eluted by imidazole, which competes with histidine-tagged proteins for column binding. Each isoform required 300 mM imidazole to elute efficiently. Then, 5 μ L of each eluted enzyme was provided with saturating levels of substrate and assessed for isopeptidase and glycosidase activity at various pH levels, which allowed the optimal pH of enzymatic activities for each isoform to be identified (Figure 1). Additional studies showed that at optimal pH, both isoforms had approximately the same k_{cat} values, but isoform 2 had a lower K_M than isoform 1 for both enzymatic activities.

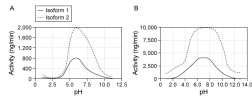


Figure 1 Isopeptidase activity (A) and glycosidase activity (B) of 5 μ L of destabilase isoforms 1 and 2 at various pH values

Isoform 2 was further characterized by mutating specific residues and measuring enzymatic activities at optimal pH. The results are shown in Figure 2 as a percentage of the activity observed in wild-type enzyme.

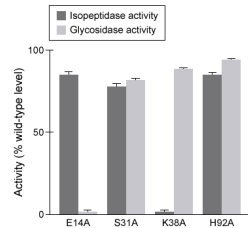
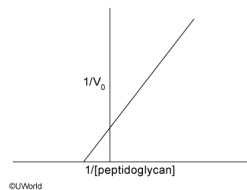


Figure 2 Isopeptidase and glycosidase activity of isoform 2 of destabilase with various mutations

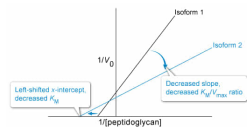


The Lineweaver-Burk plot shown above depicts the glycosidase activity of isoform 1. Based on Figure 1 and the information in the passage, which of the following would describe the appearance of the plot for isoform 2 glycosidase activity at pH 7 relative to isoform 1?

- ✓ ☒ A. The x-intercept would be left-shifted, and the slope would be decreased. [51%]
☐ B. The slope would be decreased, but the x-intercept would be unaffected. [11%]
☐ C. The x-intercept would be right-shifted, but the slope would be unaffected. [17%]
☐ D. The slope would be increased, and the x-intercept would be right-shifted. [19%]

Correct
 51% answered correctly

Explanation:



Lineweaver-Burk plots are derived from a **rearrangement** of the Michaelis-Menten equation. They are **linear graphs** that display the inverse of an enzymatic reaction rate ($1/V_0$) as a function of the inverse of substrate concentration ($1/[S]$). The **y-intercept** represents the reciprocal of the maximum reaction velocity $1/V_{max}$; the **x-intercept** represents the negative reciprocal of the enzyme's Michaelis constant $-1/K_M$; and the **slope** of the line is the ratio K_M/V_{max} .

The passage states that isoform 2 has a lower K_M than isoform 1. A decrease in K_M corresponds to a left-shift in the x-intercept because as K_M *decreases*, the magnitude of $-1/K_M$ *increases*, moving it farther away from the origin. For example, a K_M of 4 yields an x-intercept of $-1/4$, whereas a lower K_M such as 2 would yield an x-intercept of $-1/2$, which is **to the left** of $-1/4$ on the x-axis. Therefore, the x-intercept for isoform 2 would be **left-shifted** relative to isoform 1.

Figure 1 shows that at saturating substrate concentrations, isoform 2 has a higher activity (and therefore a higher V_{max}) at pH 7 than isoform 1. Because the slope of a Lineweaver-Burk plot is equal to the K_M/V_{max} ratio, a decreased value for K_M (decreased numerator) and an increased V_{max} value (increased denominator) both contribute to a decreased slope. Therefore, the plot for isoform 2 would have a **decreased slope** compared to isoform 1.

(Choice B) If the x-intercept were **unaffected**, then both isoforms would have the same K_M . The passage states that isoform 2 has a *lower* K_M than isoform 1.

(Choices C and D) A right-shift in the x-intercept would indicate an *increase* in K_M , but the passage says that K_M is *decreased* in isoform 2 compared to isoform 1. In addition, an **unaffected slope** would indicate an unchanged K_M/V_{max} ratio, and an **increased slope** would indicate an increased K_M/V_{max} ratio.

Educational objective:

Lineweaver-Burk plots display enzyme kinetics in linear form. The y-intercept corresponds to $1/V_{max}$; the x-intercept corresponds to $-1/K_M$; and the slope corresponds to K_M/V_{max} .

Time spent: 0 sec
 Subject:
 Biochemistry

QID:402747
 System:
 Enzymes

Destabilase, an enzyme found in the saliva of leeches, exhibits at least two activities: isopeptidase activity and glycosidase activity. Isopeptidase activity is believed to prevent blood clotting by hydrolyzing the isopeptide bond between the ϵ -amino group of lysine and the γ -carboxyl group of glutamate in fibrin monomers. The enzyme's glycosidase activity digests sugars found in the peptidoglycans of bacterial cell walls and may play a role in innate immunity.

Two isoforms of destabilase were tagged with a polyhistidine tail and recombinantly expressed in *E. coli* cells. Both isoforms were then purified by affinity chromatography and eluted by imidazole, which competes with histidine-tagged proteins for column binding. Each isoform required 300 mM imidazole to elute efficiently. Then, 5 μ L of each eluted enzyme was provided with saturating levels of substrate and assessed for isopeptidase and glycosidase activity at various pH levels, which allowed the optimal pH of enzymatic activities for each isoform to be identified (Figure 1). Additional studies showed that at optimal pH, both isoforms had approximately the same k_{cat} values, but isoform 2 had a lower K_M than isoform 1 for both enzymatic activities.

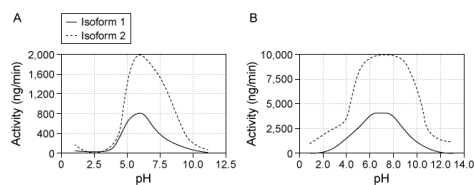


Figure 1 Isopeptidase activity (A) and glycosidase activity (B) of 5 μ L of destabilase isoforms 1 and 2 at various pH values

Isoform 2 was further characterized by mutating specific residues and measuring enzymatic activities at optimal pH. The results are shown in Figure 2 as a percentage of the activity observed in wild-type enzyme.

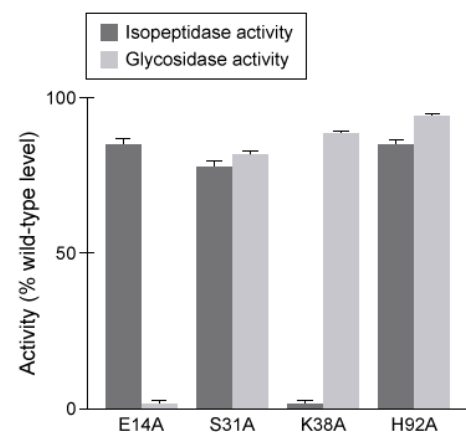


Figure 2 Isopeptidase and glycosidase activity of isoform 2 of destabilase with various mutations

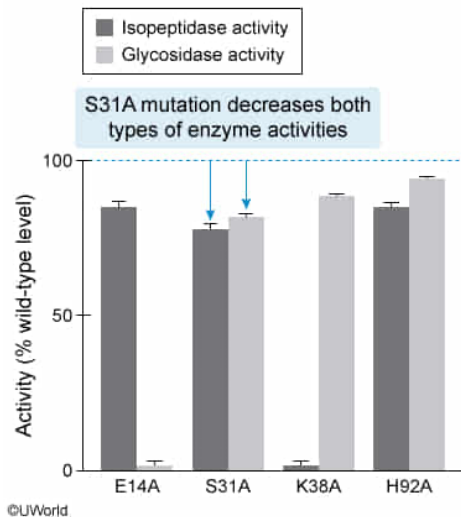
Which conclusion is LEAST supported by the data in Figure 2? (Note: Assume all differences relative to wild-type levels are statistically significant.)

- ☐ A. Glutamate at position 14 is required for glycosidase activity but not for isopeptidase activity. [6%]
- ☒ B. Mutating serine at position 31 has no effect on isopeptidase activity or glycosidase activity. [49%]
- ☐ C. Lysine at position 38 is required for isopeptidase activity but not for glycosidase activity. [6%]
- ☐ D. Mutating histidine at position 92 has the smallest impact of the tested mutations on glycosidase activity. [37%]

Correct

48% answered correctly

Explanation:



Enzymes are proteins that **catalyze** chemical reactions in biological systems. **Some amino acids** are required for **proper protein folding** or **catalysis**; mutating these amino acids will most likely **abolish enzyme activity entirely**. Other amino acids may help *facilitate* activity but are **not required** for enzyme function. Mutating these amino acids results in either **no change** or a **partial decrease** in reaction rate but does not completely eliminate activity. The effect of a particular amino acid on enzymatic activity must be **determined experimentally**.

The question asks for the conclusion that is LEAST supported by the data. Figure 2 shows the effects of several mutations on both isopeptidase and glycosidase activities in isoform 2 of the destabilase enzyme and expresses them as a percentage of the activity observed in the wild-type enzyme.

- When glutamate at position 14 is mutated to alanine (**E14A**), isopeptidase activity remains high but glycosidase activity is essentially zero. Therefore, Figure 2 *strongly* supports the conclusion that this glutamate residue is required for glycosidase activity but *not* for isopeptidase activity (**Choice A**).
- When lysine at position 38 is mutated to alanine (**K38A**), isopeptidase activity is essentially zero, whereas glycosidase activity is only slightly diminished. Therefore, Figure 2 *strongly* supports the conclusion that this lysine residue is required for isopeptidase activity but *not* for glycosidase activity (**Choice C**).
- When histidine at position 92 is mutated to alanine (**H92A**), glycosidase activity remains near 100% of wild-type levels and is higher than the glycosidase activity of any other mutation tested. Therefore, mutating this histidine residue *does* have the smallest impact of all the tested mutations on glycosidase activity (**Choice D**).

In contrast, mutating serine at position 31 (S31A) does not completely abolish either isopeptidase or glycosidase activity, but it does *diminish* both activities. Therefore, **the conclusion that this mutation has no effect** on either type of enzyme activity is **LEAST supported by the data** and correctly answers the question.

Educational objective:

Mutations to amino acids that are critical for proper folding or catalysis typically result in complete loss of enzyme activity, whereas mutations in less critical amino acids result in small changes. The effects of a mutation must be determined experimentally.

Time spent: 0 sec
Subject:
Biochemistry

QID: 402748
System:
Enzymes